| | ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University) | | Assignment |
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| Branch | **Computer Science and Engineering, Computer Science and Information Technology** | Programme | **B.Tech** |
| Course Name | **Science and Technology in Ancient India** | Semester | **VIII** |
| Course Code | **IKS4003** | Academic Year | **2026/Even** |
| **Assignment-1** | **Topic- Science and Technology- The Beginning** | **GP-1** | |

| Learning Level (LL) | | | |
|---|---|---|---|
| | **L1**: Remembering | **L3:** Applying | **L5**: Evaluating |
| | **L2**: Understanding | **L4**: Analyzing | **L6**: Creating |

| Q's | Questions | COs | LL |
|---|---|---|---|
| **1** | Why is it necessary to address the dharma paradigm using scientific language? | **CO1** | **L1** |
| **2** | What were the main features of architecture during the Harappan period? | **CO1** | **L3** |
| **3** | How was architecture in the Vedic period different from Harappan architecture? | **CO1** | **L3** |
| **4** | Briefly explain the primary cause of natural diversity according to Darwin's theory versus Buddhist philosophy. | **CO1** | **L1** |
| **5** | What is the "encephalization factor," and which being possesses the highest value? | **CO1** | **L3** |
| **6** | What are Buddhist stupas and viharas? | **CO1** | **L3** |
| **7** | What are the main styles of temple architecture in India? | **CO1** | **L3** |
| **8** | What is rock-cut architecture? | **CO1** | **L3** |
| **9.** | List the core assumptions found in Darwin's theory of evolution. | **CO2** | **L3** |
| **10** | What is the scope of Ayurveda in ancient India? | **CO2** | **L3** |

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| 11 | What is the origin and antiquity of Ayurveda? | CO1 | L1 |
| 12 | What is the fundamental difference between simply surviving nature and surmounting natural hindrances? | CO2 | L3 |
| 13 | How did Ayurveda develop and decline over time? | CO2 | L3 |
| 14 | How was Ayurveda applied to other forms of life? | CO2 | L3 |
| 15 | How did Ayurveda spread outside India? | CO2 | L2 |
| 16 | How is "Karma" defined specifically within the *Rajavartika* of the Jaina tradition? | CO2 | L3 |
| 17 | What was astronomy called in the Brahmana period? | CO2 | L2 |
| 18 | How do long-lasting desire and effort (Karma) contribute to the perceived superiority of humans among species? | CO2 | L3 |
| 19 | How was the earth described in the Satapatha Brahmana? | CO2 | L3 |
| 20 | Who is said to have calculated solar eclipses in the Rg-Veda? | CO2 | L2 |

| Assignment 1 | Topic: Science and Technology- The Beginning | Date of Assignment1: 25.02.2026 | Date of Submission: 04.03.2026 |
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**Note:**

1. assignment carries weightage of **20 marks out of 100**
2. Course outcome CO1 to CO2 was covered.

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| Course Outcomes | CO1 | To provide a comprehensive overview of the historical development of scientific thought and technological practices in ancient India, from the Vedic period to the medieval era. |
| | CO2 | To analyse ancient Indian contributions to fundamental sciences, Mathematics (*Sulba − Sutras*), Astronomy (*Jyotisha*), and Physics. |
| | CO3 | To evaluate technological advancements in fields such as Metallurgy, Civil Engineering, and Architecture (Vastu Shastra). |
| | CO4 | To examine the evolution of medicinal systems, specifically the theories and surgical practices found in the foundational texts of Ayurveda |
| | CO5 | To foster a critical understanding of how ancient Indian scientific knowledge was documented in Sanskrit texts and its subsequent influence on global scientific thought through translation and trade. |
| | CO6 | To investigate the Vedic and Post-Vedic understanding of heat transfer, latent heat, and the role of Agni in the phase transitions of matter (solid to liquid to gas). |